

Gurugram University

Scheme of Studies and Examination

B.TECH (Civil Engineering) – 6th Semester

Sr. No.	Course Code	Course Title	Hours per Week	Contact Hours per Week	Credit	Examination Schedule (Marks)				Duration of Exam (Hours)
			L-T-P			Class work	Theory	Practical	Total	
1.		Irrigation And Hydraulic Structures	3-0-0	3	3	30	70	-	100	3
2.		Foundation Engineering	3-0-0	3	3	30	70	-	100	3
3.		Estimation ,Costing and Valuation	3-0-0	3	3	30	70	-	100	3
4.		Professional Elective-I	3-0-0	3	3	30	70	-	100	3
5.		Professional Elective-II	3-0-0	3	3	30	70	-	100	3
6.		Open Elective-I	3-0-0	3	3	30	70	-	100	3
7.		Environmental Engineering Lab.	0-0-2	2	1	50	-	50	100	3
8.		Foundation Engineering lab	0-0-2	2	1	50	-	50	100	3
9.		Irrigation Drawing LAB	0-0-2	2	1	50	-	50	100	3
10.		Estimation,Costing and Valuation LAB	0-0-2	2	1	50	-	50	100	3
11.		Computer aided Civil Engineering Design	0-0-2	2	1	50	-	50	100	3
12.		Seminar-I	0-0-2	2	1	-	-	50	50	3
		TOTAL			24				1150	

	Course Code	Course Title
**Elective –I		1. Waste Water Treatment
		2. Advanced Concrete Structure

	Course Code	Course Title
**Elective –II		1. Pre-Stressed Concrete
		2. Repair & Rehabilitation Of Structure

	Course Code	Course Title
**Open Elective –I		3. Environmental impact assessment and life cycle analyses
		4. Earthquake Engineering

Note:

2. Each student has to undergo practical training of 6 weeks during summer vacation after 6th semester

and its evaluation shall be carried out in 7th Semester.

IRRIGATION AND HYDRAULIC STRUCTURE			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

Course Objectives

- To make the students conversant with introduction to irrigation and national policies.
- To understand the basic methods of irrigation and soil water relation.
- To expose the students to water logging and land reclamation.
- To provide adequate knowledge regarding river training and canal outlet.
- To have adequate knowledge of drainage work.

COURSE CONTENT

Unit-I

Module 1: Irrigation and Water Requirement of Crops

Scope and Necessity of Irrigation, Different Water Resources, Development of Irrigation in India, Benefits of Irrigation, National Water Policy

Crops and Crop Seasons in India, Field Capacity, Wilting Point, Duty and Delta, Soil-Water Relationship- root zone soil water, infiltration, frequency of irrigation; Methods of Irrigation

Module 2: Canal Regulation and Cross Drainage Works

Canal Fall- Necessity and Location, Types of fall, Design of fall , Cross Regulator and Distributory Head Regulators, Silt Control Devices, Canal Escapes and its Types.

Cross Drainage Works - Classification, Site Selection Criteria, Factors Affecting the Selection of Cross Drainage Works, Hydraulic Design of- aqueducts, syphon aqueducts, super passage, canal syphon and level crossing.

Unit-II

Module 3: Canal Outlets

Essential Requirements for an Outlet, Classification and Types of Outlets, Salient Features and Design: Pipe Outlet, APM Outlet and Open Flume Outlet; Flexibility Proportionality, Setting and Sensitivity of Outlet.

Module 4: Spillways and Energy Dissipations

Essential Requirements of Spillway and Spillway's Capacity, Types of Spillways and their Suitability, Ogee Spillways, Chute, Side Channel, Shaft and Syphon Spillways, Energy Dissipaters and its types ; Stilling Basins, USBR and IS Stilling Basins.

Unit-III

Module 5: River Training

Objectives and Classification of River Training Works, Methods and Planning of River Training, Marginal Embankments, Guide Banks, Spurs, Cut Offs, Bank Protection and Launching Apron.

Module 6: Water Logging and Drainage

Water Logging- Effects, Causes & Preventive Measures, Land Drainage, Essential Requirements of a Drain, Classification of Drains, its Operation and Maintenance.

Unit-IV

Module 7:Dams

Planning and investigations of reservoir and dam sites, Choice of dams, Dam construction problems, Forces acting on gravity dams, modes of failure and design criteria for structural stability of gravity dams, seepage line in a homogenous earth dam, equipotential lines, path lines.

Module 8: Hydraulic structure

Stream lines, critical exit gradient, need of weirs and barrage, Factors controlling the design of weirs and barrages and its components. Design of weirs.

Course Outcomes

At the end of the course, the students will be able to:

- Learn historical development of irrigation in India and the policies framed
- Learn about various methods of irrigation
- Understand water logging effects and methods of land reclamation
- Know about river training, classification and requirement of canal outlets and cross drainage works.

References:

- Garg, S. K., "Irrigation Water Power & Water Resources Engg." Standard Publishers & Distributors, Delhi, latest edition
- Modi, P.N. "Irrigation, Water Resources and Water Power Engg." Standard Book House, N. Delhi latest edition
- Arora, K R "Irrigation Water Power & Water Resources Engg." Standard Publishers & Distributors, Delhi, latest edition
- Sharma, S.K., Principles and Practice of Irrigation Engineering, S.Chand & Co., latest edition
- Punmia, B.C., "Irrigation and Water Power Engg." Standard Publishers, latest edition

FOUNDATION ENGINEERING			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

Course Objectives

- To emphasize the importance of soil investigations.
- To provide the knowledge for carrying out field investigations and to identify soils in geotechnical engineering practice.
- To identify factors controlling soil behaviour and methods of dewatering.
- To understand different types of foundations and their importance in field.
- To identify different parameters for determining the bearing capacity of soil.
- To explain under which conditions deep foundation is needed and how to estimate pile and pile group capacity.
- To understand dynamic loading on soil foundation system and provide knowledge to lay out caissons and well foundation.

COURSE CONTENT

Unit-I

Module 1: Sub-Surface Exploration

Scope and Objectives, Stages in Soil Exploration, Depth and Lateral Extent of Exploration, Guidelines for Various Types of Structures, Ground Water Observations, Methods of Bore Holes, Soil Sampling and Disturbance, Major Types of Samplers, Sounding Methods-SPT, SCPT, DCPT and their Interpretation, Geophysical Methods, Pressure-Meter Test, Exploration Logs.

Module2: Drainage and Dewatering

Ditches and Sumps, Well Point Systems, Shallow Well System, Deep Well Drainage, Vacuum Method, Electro-Osmosis, Consolidation by Sand Piles.

Unit-II

Module 3: Shallow Foundations

Types of Foundations, Depth of Foundation, Types of Shallow Foundations and their Relative Merits, Design Criteria for Structural Safety of Foundation: i) Location of Footing, ii) Shear Failure Criterion, iii) Settlement Criterion. Modes of Shear Failure, Rankine's Analysis Tergazi's Theory, Skempton's Formula, Meyerhoff's Bearing Capacity Theory, Effect of G.W.T. , Effect of Eccentricity on Bearing Capacity, Inclined Load, IS Code Recommendations.

Module 4: Settlement of Foundations

Various Causes of Settlement of Foundation, Allowable Bearing Pressure Based on Settlement, Elastic and Consolidation Settlement, Allowable Settlement According to IS Code Method, Plate Load Test and its Interpretation, Conventional Procedure of Proportioning of Footings, Situation Suitable for the Shallow Foundations.

Unit-III

Module 5: Bearing Capacity of foundations

Design Bearing Capacity, Bearing Capacity from Penetration Tests, Factors Affecting Bearing Capacity, Methods of Improving Bearing Capacity, Raft Foundations, Bearing Capacity of Raft in Sands and Clays, Various Methods of Designing Rafts, Seismic Considerations, Floating Foundations.

Module 6: Pile Foundations

Necessity of Pile Foundations, Classification of Piles, Selection Criteria, Load Capacity, Static Analysis, Analysis of Pile Capacity in Sands and Clays, Dynamic Analysis, Pile Load Tests, Negative Skin Friction, Batter Piles, Lateral Load Capacity, Uplift Capacity of Single Pile, Under-Reamed Pile, Batter Pile. Group Action in Piles, Pile Spacing, Pile Group Capacity, Stress on Lower Strata, Settlement Analysis, and Design of Pile Caps.

Unit-IV

Module 7: Drilled Piers and Caisson Foundations

Drilled Piers- Types, Uses, Bearing Capacity, Settlement and Construction Procedure; Caissons- Types, Bearing Capacity, Settlement and Construction Procedure

Module 8: Well Foundations

Shapes, Depth of Well Foundations, Components, Factors Affecting Well Foundation Design Lateral Stability, Construction Procedure, Sinking of Wells, Rectification of Tilts and Shifts, Recommended Values of Tilts & Shifts as per IS: 3955

Course Outcomes

At the end of the course, the students will be able to:

- To understand the importance of soil investigation and carry out sub-surface explorations for any civil engineering construction.
- To evaluate Bearing capacity factors and estimate bearing capacity using suitable methods.
- To do proper foundation proportioning for any kind of shallow foundation system.
- To estimate pile and pile group capacity for any kind of soils including group efficiency.
- To determine safe bearing capacity for various foundation system by considering shear and settlement criterion.

References:

- Murthy, V.N.S, A text book of Soil Mechanics and Foundation Engineering, UBS Publishers & Distributors Pvt. Ltd., New Delhi, latest edition
- Punmia, B.C., Soil Mechanics and Foundation Engineering, Laxmi Publications Pvt. Ltd., New Delhi, latest edition
- Gopal Ranjan, ASR Rao, Basic and Applied Soil Mechanics, New Age International Ltd. Publishers- N.Delhi, latest edition
- Nainan P Kurian, Design of foundation Systems Principles and Practices, Narosa, latest edition
- Braja M. Das, Principles of Foundation Engineering, Thomson Asia Pvt. Ltd., Singapore, latest edition
- Donald P. Coduto, Man-Chu Ronald Yeung and William A. Kitch, Geotechnical Engineering, Principles and Practices, PHI Learning Private limited, latest edition
- P.Purshotam Raj, Geotechnical Engg, Tata Mcgraw Hill, N.Delhi, latest edition

Course code					
Category	Professional Elective course				
Course title	Estimation, Costing and Valuation				
Scheme and Credits	L	T	P	Credits	Semester 6 th
	2	1	0	3	
Class work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3 Hours				

Course Objectives:

The objective of this Course is

- To analyze cost/revenue data and carry out make economic analyses in the decision making process to justify or reject alternatives/projects on an economic basis.
- Prepare engineering students to obtain professional licensure.
- To function in the business and management side of professional engineering practice.
- To preparation estimate of the civil engineering works.
- To preparation specification of construction items.

COURSE CONTENT

Unit-I

Module 1: Introduction

Purpose of estimating and valuation, Principle of estimation, unit of measurement, item work, Different kinds of estimates, Different methods of estimation. Introduction to estimates of other Civil engineering structures.

Module 2: Building Estimate

Estimation of quantity of load bearing structure with single room & two rooms, Estimation of quantity single storied residential building. Estimation of materials in multi-storey buildings with different sections of walls, foundation, floors and roofs, reinforced brick work, R.C.C works, finishing works and Lump sum items, Estimates of canals, dams, barrages and other structures.

Unit-II

Module 3: Specification

Objectives and Necessity of specification, Types of specification, General specification, Specification of different construction materials, Specification as per building classification, Language of specific writing. Specification of Works: Detailed specification for earthworks, cement, concrete, brickwork, flooring, D.P.C, R.C.C, cement plastering, painting and other finishing.

Module 4:Market Survey

Traditional and modular materials, Market survey of construction materials, Wages of labour, Tool's plant and equipment of construction.

Unit-III

Module 5: Rate Analysis

Prerequisites, factors affecting rate analysis, over head expenses, Procedure for rate analysis: schedule of rates, labour requirement for different works, material requirement for different works, Rate analysis of different Items of work (Earth work, Concrete works, R.C.C works, Reinforce Brick work, plastering, painting, finishing).

Module 6:Abstracting and Billing

Purpose of abstract, preparation of abstract, measurement book, cash book, preparation, examination and payment of bills, first and final bills, administrative sanction, technical sanction. Billing – maintenance of muster roll,record of bills, vouchers and receipt book

Unit-IV

Module 7: Tendersand Contracts

Tender notice and documents, acceptance of tender, Earnest money, security money, retention money, Contract-contractor: terms and conditions of contract, Agreement, Form of Contract, Responsibility of owner, Architect, Contractor and Engineer.Preparation of pay bill, measurement of work for payment of contractors, different types of payment – first &final, running advance and final payment.

Module 8: Valuation

Purpose of valuation, principles of valuation,Types of property, Depreciation, Sinking fund, Lease hold and free hold property, obsolescence, Gross income, Outgoing and Net income, Capitalized value and year's purchase. valuation of a building – cost method, rental – return method.

Course Outcomes:

At the end of the course, students shall be able:

- To understand the methodology of Cost-driven design optimization.
- To get understanding of contract models, contract plans and specifications.
- The students will learn the purpose and importance of valuation

Referenced Books:

1. Dutta BN – Estimating &costing.
2. Chakraborty – Estimate costing &specification in civil engg.
3. Kohli &kohli – A text book on estimating &costing (Civil) with drawings Ambala ramesh Publications.

WASTE WATER TREATMENT			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

Course Objectives

- To learn basics of sewage collection and design of sewers
- To learn the basics of sewage composition and its characteristics
- To have adequate knowledge about various sewage treatment processes and its design
- To provide adequate information on various disposal standards for treated effluents

COURSE CONTENT

Unit-I

Module 1: Introduction

Importance and Necessity of Sanitation, Terms used in sanitation- Sullage, Sewage, Sewer and Sewerage, Sewerage Systems and their Suitability, Dry Weather Flow, Factors Affecting Dry Weather Flow, Flow Variations and their Effects on Design of Sewerage System.

Module 2: Design of Sewers

Types of Sewers and their Hydraulic Design, Material used for Sewer Construction, Joints and Sewer Appurtenances, Layout, Construction and Testing of Sewer Lines, Velocity in Sewers, Storm Water Sewers

Unit-II

Module 3: House Drainage

Principles of House Drainage, Types of Pipes used for Drainage , Classification and Functions of Traps, Sanitary Fitting, System of Plumbing, House Drainage Plan and Ventilation of House Drainage.

Module 4: Wastewater Characteristics

Quality Parameters- Physical, Chemical and Biological Characteristics, Oxygen Demand, Indian Standards for Disposal of Effluents into Inland Surface Sources and on Land, Guidelines for Reuse of Treated Wastewater

Unit-III

Module 5: Primary Sewage Treatment

Objectives, Flow chart of Conventional Treatment Units and their Efficiencies, Preliminary Treatment, Screening and Grit Removal Units, Principle, Types and Design of Primary Sedimentation Tank, Coagulation Aided Sedimentation Tank, Flocculation,

Module 6: Secondary Sewage Treatment

Concept of Organic Matter Removal, Aerobic and Anaerobic Treatment Processes, Activated Sludge Process, Conventional and Extended Aeration Systems, Trickling Filters, Aerated Lagoons, septic tank, Waste Stabilization Ponds, Oxidation Ditches, Up-Flow Anaerobic Sludge Blanket Process.

Unit-IV

Module 7: Sludge Treatment

Objectives, Sludge Digestion, Digestion and Disposal of Primary and Secondary Sludge, Factors Affecting Sludge Digestion, Thickening of Sludge, Anaerobic Digestion of Sludge, Sludge Digestion Tank, Sludge Conditioning and Dewatering, Sludge Drying Bed

Module 8: Sludge Disposal

Standards of Wastewater Disposal, Modes of Disposal of Treated Sludge, Self-Purification of Streams, Oxygen Sag Curve, Sewage Farming, Sodium Hazards, Soil Dispersion System

Course Outcomes

At the end of the course, the students will be able to:

- Estimate quantity of sewage and design sewerage system
- Determine the various characteristics of sewage
- Design various sewage treatment units
- Plan reuse of treated effluent and select appropriate disposal option

References:

- Environmental Engineering: Peavy H. S., Rowe D. R. and Tchobanoglous G.
- Wastewater Engineering, Collection, Treatment and Disposal: Metcalf and Eddy
- Water Supply and Sanitary Engineering: Birdie, G. S. and Birdie
- Sewage and Sewage Treatment: S.K. Garg.
- Sewage and Sewage Treatment: S.R. Krishansagar.
- Waste Water Engineering: B.C. Punmia.
- Manual on Sewerage and Sewage Treatment: Ministry of Urban Dev., New Delhi

ADVANCED CONCRETE STRUCTURE			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

Course Objectives

At the end of this course, the student should be able to impart understanding of designing specialized RCC structures. To prepare the detailed structural drawings for execution purpose

COURSE CONTENT

Unit-I

Module1: Continuous Beams

Basic Assumptions, Moment of inertia, Settlements, Modification of Moments, Maximum Moments and Shear, Design Examples.

Module2: Curved Beams

Analysis for Torsion, Redistribution of Moments for Single and Multi-Span Beams, Design of Circular Beams Supported on Symmetrically Placed Columns, Semi-Circular Beams Supported on Equally Spaced Column , Design examples.

Unit-II

Module 3: Flat Slab

Advantages of Flat Slab, General Design Considerations, Indian Code Recommendations, Approximate Direct Design Method, Equivalent Frame method, Design of Flat Slabs, Openings in Flat Slab

Module 4: Yield Line Theory

Basic Assumptions, Yield Line Patterns and Failure Mechanisms, Ultimate Load on Slab, Design Example.

Unit-III

Module 5: Liquid Retaining Structure

Design Concepts of Liquid Retaining Structures, Design of Tanks Resting on Ground, Underground Tanks and Overhead Service Reservoirs, Staging and Foundation Design.

Module 6: Stair Case

Various Types of Staircases, General Notes on Design of Stair, Design Examples.

Unit-IV

Module 7: Design of Joints

Types of Joints, Joints in Multi-Storied Buildings, Forces Acting on Joints, Design of Joints for Strength, Anchorage Requirement in Joints, Detailing of Reinforcement in Joints.

Module 8: Building Frames

Introduction, Members Stiffness, Torsion in Buildings, Design Loads on Building Frames Including Wind and Earthquake Loads, Earthquake Resistant Design using Software, Introduction to IS: 13920 and Concepts of Ductile Detailing in Building Frames, Design and Detailing for Ductility, Design Examples.

Course Outcomes

At the end of the course, the students will be able to:

- Design advanced RCC structures.
- Prepare detailed structural drawings for the designed RCC structures using software.

References:

- “Advanced Reinforced Concrete Design”, P.C. Varghese, Prentice Hall of India Pvt. Ltd.
- “Plain & Reinforced Concrete,” Jain & Jai Krishan (Vol. I & Vol-II), Nem Chand and Bros.
- “Reinforced Concrete Structures”, Syal and Goel, S. Chand & Company Pvt. Ltd.
- “Reinforced Concrete Design”, S.U. Pillai & Devdas Menon, Tata McGraw Hill.
- "Reinforced Concrete Limit State Design" A.K. Jain, Nem Chand and Bros.

REPAIR & REHABILITATION OF STRUCTURE			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
*_		Duration of Examination:	3 hrs

Course Objectives:

This course has been designed with an aim to give the students an insight into the subject of concrete repair, its protection and strengthening

COURSE CONTENT

Unit-I

Module 1: Introduction

Overview of Distress, Deterioration in Concrete Structures, Global Scenario of Distressed Structures, Need for Repairs and Upgrading of Structures, Process for Durable Concrete Repair

Module 2: Deterioration of concrete structures

Types of Deterioration - causes & symptoms, mechanism & micro-structure of concrete, Physical and Chemical Deterioration and its factors, Deterioration due to Water Leakage, Fire – detection & mitigation, Deterioration due to Ageing and Inadequate Maintenance, Design & Construction Deficiencies like Overloading.

Unit-II

Module 3: Visual deterioration of structures

Types of Cracks, Causes & Characteristic of Cracking in Various Structural Components, Measurement of Cracks and Interpretation of the Cracking Phenomena

Module 4: Conditional/damage assessment & Evaluation of structures

Structural Assessment- importance, objective, various stages, conditional evaluation of the structure, Damage Assessment Procedure, Preliminary & Detailed Investigation – scope, objectives, methodology & rapid visual inspection of structures , Damage Assessment Allied Tests (Destructive, Semi-Destructive and Non-Destructive), Field & Laboratory Testing Procedures- strength, corrosion activity, performance & Integrity, Durability.

Unit-III

Module 5: Repairs of concrete structures

Repairing Materials- criteria, selection of repair materials, methodology, performance requirements, preparatory stage of repairs, different types of repair materials & their application and repair techniques

Module 6: Retrofitting/Strengthening

Need for Retrofitting, Design Philosophy of Strengthening Structures, Conventional and Advanced Techniques Available for Strengthening, Seismic Retrofit of Concrete Structures- deficiencies in

structure requiring seismic retrofit and its design philosophy, Latest Techniques to Enhance the Seismic Resistance of Structures.

Unit-IV

Module 7: Protection & maintenance of structures

Importance of Protection & Maintenance, Categories of Maintenance, Building Maintenance, Corrosion Mitigation Techniques

Module 8: Structural health monitoring (SHM)

Definition and Motivation for SHM, Basic Components of SHM and its Working Mechanism, SHM as a Tool for Proactive Maintenance of Structures

Course Outcomes

At the end of the course, the students will be able to:

- Identify and define all the terms and concepts associated with deterioration of concrete structures.
- Carry out the damage assessment and Rapid Visual inspection of a building showing signs of deterioration and thus should be able to detect the possible cause /source of deterioration.
- Develop a knowhow of the Concrete repair industry equipped with variety of repair materials and techniques.
- Describe and apply the importance of quality control in concrete construction and significance of protection and maintenance of structures.

References:

- Concrete microstructure, Properties and materials – P Kumar Mehta and Paulo J.M.Monterio.
- Handbook on Repairs and Rehabilitation of RCC buildings – CPWD, Government of India.
- Concrete Technology by M.L.Gambhir, Tata McGraw-Hill Education, Third Edition
- V. M. Malhotra, Nicholas J. Carino 2004 “Handbook on Nondestructive Testing of Concrete”
- “Repair and Strengthening of Concrete structures” , FIP guide, Thomas Telford, London.
- Concrete Structures, Protection, Repair and Rehabilitation by R.Dodge Woodson.
- Repairs and rehabilitation of concrete structures by P. I. Modi & C. N. Patel, PHI Publication.

PRE-STRESSED CONCRETE			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

Course Objectives

- To introduce the need for prestressing as well as the methods, types and advantages of prestressing to the students.
- Students will be introduced to the design of pre-stressed concrete structures subjected to flexure and shear.
- To make them familiar with design of typical pre-stressed concrete structural elements and to have a knowledge of the codal provisions.

COURSE CONTENT

Unit-I

Module 1: Material Properties and Prestressing Systems

Introduction of Prestressing system- history, types, source, advantages and its limitations, Prestressing Systems and Devices

Materials properties- Constituents of Concrete and their Properties, Stress-Strain Curves for Concrete, Constituents of Prestressing Steel and their Properties, Types of Prestressing Steel, Stress-Strain Curves for Prestressing Steel, Relaxation of Steel, Fatigue, Codal Provisions.

Module 2: Losses in Prestress

Elastic Shortening, Pre-Tensioned & Post-Tensioned - Axial & Bending Members, Losses in Prestress - Friction, Anchorage Slip, Creep of Concrete, Shrinkage of Concrete, Relaxation of Steel, Total Time-Dependent Loss, Force Variation Diagram.

Unit-II

Module 3: Analysis and Design for Shear and Torsion

Analysis for Shear- Introduction, Types of Cracks, Components of Shear Resistance, Modes of Failure, Effect of Prestressing Force, Stress in an Uncracked Beam, Design and Detailing Requirement for Shear, Design of Transverse Reinforcement.

Analysis for Torsion-Introduction, Crack Pattern Under Pure Torsion, Components of Resistance for Pure Torsion, Modes of Failure, Effect of Prestressing Force for Torsion, Stresses in an Uncracked Beam, Design and Detailing Requirement for Torsion, Design of Longitudinal Reinforcement and Transverse Reinforcement.

Module 4: Calculations of Deflection and Crack Width

Factors Influencing Deflections, Short Term Deflections of Uncracked Members, Prediction of Long Term Deflections due to Creep and Shrinkage, Check for Serviceability Limit State of Deflection. Deflection due to Gravity Loads and Prestressing Force, Total Deflection, Determination of Moment of Inertia, Calculation of Crack Width, Method of Calculation, Limits of Crack Width.

Unit-III

Module 5: Analysis of Members

Analysis of Members under Axial and Flexural Load- Based on Stress, Force and Load Balancing Concept. Cracking Moment, Kern Point, Pressure Line, Analysis for Ultimate Strength, Variation of Stress in Steel Condition at Ultimate Limit State, Analysis of Rectangular Sections, Flanged Sections, Partially Pre-Stressed Sections, Un-Bonded Post-Tensioned Beams.

Module 6: Design of Members

Design of Members for Axial Tension, Flexure Type I and Type II, Choice of Sections, Determination of Limiting Zone, Post-Tensioning in Stages, Magnel's Graphical Method, Guyon's Method

Unit-IV

Module 7: Composite and Continuous Beams

Analysis and Design of Composite Beams – Methods of Achieving Continuity in Continuous Beams, Analysis for Secondary Moments, Concordant Cable and Linear Transformation, Calculation of Stresses, Principles of Design.

Module 8: Miscellaneous Structures

Design of Tension and Compression Members, Tanks, Pipes and Poles, Partial prestressing – definition, methods of achieving partial prestressing, merits and demerits of partial prestressing.

Course Outcomes

At the end of the course, the students will be able to:

- Analyse prestressed concrete members
- Design prestressed concrete members using codal provisions
- Design for shear and torsion of prestressed concrete members
- Design end blocks and provide detailing of reinforcements
- Design composite members and other applications
- Design continuous members

References:

- Rajagopalan.N, “Prestressed Concrete”, Narosa Publishing House, latest edition
- Dayaratnam.P., “Prestressed Concrete Structures”, Oxford and IBH, latest edition
- Lin T.Y. and Ned.H.Burns, “Design of prestressed Concrete Structures”, Third Edition, Wiley India Pvt. Ltd., New Delhi, latest edition
- IS1343:1980, Code of Practice for Prestressed Concrete, Bureau of Indian Standards,

ENVIRONMENTAL IMPACT ASSESSMENT AND LIFE CYCLE ANALYSES			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
OPEN ELECTIVE-I		Duration of Examination:	3 hrs

Course Objectives

- To understand and evaluate the impact of any activity (large or small scale) on the surrounding environment
- To introduce about mitigation strategies to protect the environment leading to sustainability
- To understand the intricacies of Life Cycle Analysis and apply basic knowledge for coherent existence

UNIT-I

Module 1

Introduction Definition, Objective with legal aspect of Environmental Impact Assessment (EIA)

Module 2

Methodology for EIA with Base Line Studies, Screening , Scoping and Public Consultation

UNIT-II

Module 3

EIA Analysis Data Collection & Environmental Impact Analysis, preparation of EIA report

Module 4

EIA Mitigation and Audit- Mitigation and Impact Management with various case studies, Environmental Audit

UNIT-III

Module 5

Introduction to Life Cycle Analysis (LCA): History, Definition, Standards and structure of LCA Goal and Scope of LCA: System of a product with boundary, unit process and functional unit

Module 6

Life Cycle Interpretation and Inventory: Limitation of LCA, Identification of significant issues, Evaluation, Reporting, Critical Review. Inventory: Data Collection, Data Bases, Allocation, Validation

UNIT-IV

Module 7

LCA Impact Assessment and Practice: Categories, Classification, Normalization, LCA Management, Life Cycle thinking, Sustainability

Module 8

Introduction: Definition, Objective with legal aspect of Environmental Impact Assessment (EIA)

Course Outcomes

At the end of the course, the students will be able to:

- To understand and evaluate the impact of any activity (large or small scale) on the surrounding environment
- To be able to formulate mitigation strategies to protect the environment leading to sustainability
- To be able to understand the intricacies of Life Cycle Analysis and apply basic knowledge for coherent existence

Reference Book

- 1 Environmental Impact Assessment R. R. Barthwal, New Age International Publication
- 2 Environmental Impact Assessment Canter McGraw Hill Publications
- 3 Environmental Impact Assessment: Theory and Practice M. Anji Reddy B. S. Publication
- 4 Environmental Impact Assessment: Theory and Practice Peter Wathern CRC Press
- 5 Life Cycle Assessment (LCA): A Guide to Best Practice Walter Klöpffer , Birgit Grahl Wiley Publishers
- 6 Environmental Life Cycle Assessment Olivier Jolliet, Myriam Saade-Sbeih, Shanna Shaked, Alexandre Jolliet, Pierre Crettaz, CRC Press
- 7 Life Cycle Student Handbook Mary Ann Curran, Scrivener Publishing, Wiley

EARTHQUAKE ENGINEERING			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
OPEN ELECTIVE-II		Duration of Examination:	3 hrs

Course Objectives

- To provide a coherent development to the students for the courses in sector of earthquake engineering.
- To present the foundations of many basic engineering concepts related earthquake Engineering
- To give an experience in the implementation of engineering concepts which are applied in field of earthquake engineering
- To involve the application of scientific and technological principles of planning, analysis, design of buildings according to earthquake design philosophy.

UNIT-I

Module 1

Seismology: Earth's Interior and Plate Tectonics; Causes of Earthquakes and Seismic Waves; Measurement of Earthquakes and Measurement parameters;
Modification of Earthquake due to the Nature of Soil; Seismic Hazard Analysis

Module 2

Earthquake Inputs: Time History Records and Frequency Contents of Ground Motion; Power Spectral Density Function of Ground Motion; Concept of Response Spectrums of Earthquake; Combined D-V-A Spectrum and Construction of Design Spectrum; Site Specific, Probabilistic and Uniform Hazard Spectrums; Predictive Relationships for earthquake parameters;

UNIT-II

Module 3

Dynamics for Earthquake Analysis: Equations of Motion for SDOF and MDOF Systems; Undamped Free Vibration of SDOF and MDOF Systems; Mode Shapes and Frequencies of MDOF System; Rayleigh Damping Matrix; Direct Time Domain Analysis of MDOF System; Direct Frequency Domain Analysis of MDOF System; Modal Analysis in Time and Frequency Domain

Module 4

Response Analysis for Specific Ground Motion: Equations of Motion for Single and Multi- Support Excitations and Solutions; Equations of Motion in State Space and Solutions; Computational Steps for the Solutions using MATLAB; Time History Analysis of 3D Tall Buildings.

UNIT-III

Module 5

Response Spectrum Method of Analysis: Concept of Equivalent Lateral Force for Earthquake; Modal Combination Rules; Response Spectrum Method of Analysis of Structures and Codal Provisions; Response Spectrum Method of Analysis for Torsionally Coupled Systems; Response Spectrum Method of Analysis for Non-Classically Damped Systems;

Module 6

Seismic Soil - Structure Interaction: Fundamentals of Seismic Soil-Structure Interaction; Direct Method of

UNIT-IV

Module 7

Inelastic Response of Structures for Earthquake Forces; Fundamental Concepts of Inelastic Response Analysis for Earthquake Forces; Solutions of Incremental Equations of Motions for SDOF Systems; Solutions of Incremental Equations of Motions for MDOF Systems; Push over Analysis; Concepts of Ductility and Inelastic Spectrum;

Module 8

Base isolation for earthquake resistant design of structures: Base isolation concept, isolation systems and their modelling; linear theory of base isolation; stability of elastomeric bearings; codal provisions for seismic isolation, practical applications.

IS Codes Used

1. IS1893: Part I (2016),
2. IS 13920: 2016
3. IS 4326

Course Outcomes

At the end of the course, the students will be able to:

- To provide a coherent development to the students for the courses in sector of earthquake engineering.
- To present the foundations of many basic engineering concepts related earthquake Engineering
- To give an experience in the implementation of engineering concepts which are applied in field of earthquake engineering
- To involve the application of scientific and technological principles of planning, analysis, design of buildings according to earthquake design philosophy.

Reference Book Name

- 1 Earthquake resistant design of Structures Agarwal and Shrikhande PHI
- 2 Earthquake-resistant design of structures S.K. Duggal, Oxford University Press.
- 3 Elements of Earthquake Engineering Jai Krishna, A. R. Chandrashekhar and Brijesh Chandra South Asian Publishers
- 4 Earthquake Resistant Design D. J. Dowrick John Willey & Sons

ENVIRONMENTAL ENGINEERING LAB			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100

Course Objectives

- To quantify the water and wastewater pollutant.
- To analyze the physical characteristics of water and wastewater.
- To analyze the chemical characteristics of water and wastewater

List of Experiments

1. Determination of Turbidity of water by using suitable method
2. Determination of pH of given water sample.
3. Determination of Hardness of given water sample.
4. Determination of Residual Chlorine in given sample of water
5. Determination of Total Suspended and Dissolved Solids in given water sample.
6. Determination of Bio –chemical oxygen demand of waste water sample.
7. Determination of chemical oxygen demand of waste water sample.
8. Determination of Conductivity of given water sample.
9. Determination of Chlorides of given water sample
10. Determination of Alkalinity and Acidity of a given water sample.
11. Determination of Dissolved Oxygen of given waste water sample.

Course Outcomes

At the end of the course, the students will be able to:

- Quantify the water and wastewater pollutant.
- Estimate the physical characteristics of water and wastewater.
- Analyze the chemical characteristics of water and wastewater

References:

- Lab Manual, ISO 14001 Environmental Management, Regulatory Standards for Drinking Water and Sewage disposal.
- Clair Sawyer and Perry McCarty and Gene Parkin, “Chemistry for Environmental Engineering and Science”, McGraw-Hill Series in Civil and Environmental Engineering.
- Guide manual: Water & wastewater analysis, Central Pollution Control Board, Govt. of India.
- APHA standard methods for the examination of water and wastewater
- Water supply engineering, S.K. Garg

FOUNDATION ENGINEERING LAB			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100

Course Objectives

- To aware the students about hydrometer method.
- To estimate the relative density and maximum dry density of soils.
- To aware the importance of SPT, consolidometer and Triaxial test for selection of foundation.
- To estimate shear strength parameters of soil by Triaxial shear test.
- To estimate consolidation parameters of clayey soil.
- To aware about the importance of sampling and aware about the significance of plate load test.

List of Experiments

1. To determine grain size analysis using Hydrometer method.
2. To determine relative density of granular Soils.
3. To determine shrinkage limit of fine grained soil
4. To determine shear strength properties for consolidated drained conditions using Triaxial test.
5. To determine shear strength properties for consolidated undrained condition using Triaxial test.
6. To determine consolidation parameters using consolidometer.
7. To determine bearing capacity parameters using Standard Penetration Test.
8. Demonstration of Undisturbed Sampling.
9. Demonstration of cone penetration test.
10. To study of Model Plate Load Test.

Course outcomes

At the end of the course, the students will be able to:

- Classify soil using hydrometer method and estimate relative density of soils.
- To perform Triaxial test under different conditions.
- To understand the procedure and calculations of SPT and prepare soil investigation report.
- Carry out interpolation tests to determine consolidation and estimate shear strength parameters.
- To obtain soil sampling by suitable method.
- Determine essential parameters of plate load test.

References:

- Gopal Ranjan, ASR Rao, Basic and Applied Soil Mechanics, New Age International (P) Ltd. Publishers- N.Delhi, latest edition
- P. Purshotam Raj, Geotechnical Engg, Tata Mcgraw Hill, N.Delhi, latest edition

Course code					
Category	Professional Elective Course				
Course title	Estimation, Costing and Valuation Lab				
Scheme and Credits	L	T	P	Credits	Semester 6 th
	0	0	2	1	
External marks:	50 Marks				
Internal marks:	50 Marks				
Total	100 Marks				
Duration of Exam	3 Hours				

Course Objectives:

The subject aims to provide the student with:

1. An introduction to quantity surveying
2. The capability to know analysis and schedule of rates
3. The ability to know specification of materials
4. An understanding about specification of works
5. The introduction to valuation

COURSE CONTENT

1. Prepare the list of items to be executed with units for detailed estimate of a given structure from the given drawing.
2. Prepare a report on market rates for given material, labour wages, hire charges of tools & equipments required to construct the given structure.
3. Study of items with specification given in the HSR (for any ten items).
4. Recording in Measurement Book (MB) for any four items.
5. Prepare bill of quantities of given item from actual measurements (any four items).
6. Calculate the reinforcement quantities from the given set of drawings for a room size of 3m X 4 m with bar bending schedule (footing, column, beam, lintel with chajja, slab).
7. Calculate the quantity of items of work from the given set of drawings using standard measurement sheet for load bearing residential structure using description of item from DSR (1BHK Building with staircase).
8. Use the relevant software to prepare detailed estimate of a residential building.

COMPUTER-AIDED CIVIL ENGINEERING DESIGN			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100

Course Objectives

- To develop, analyze and design the various structural members in the fields of Civil Engineering using AutoCAD, STAAD Pro.
- To understand the design of structures using IS codes
- To provide the knowledge about methods of analysis and design of RCC and steel frames.

List of Experiments

1. To study the commands used in plans of buildings using AutoCAD.
2. To prepare 2 D architectural drawing
3. To prepare 3 D architectural drawing
4. To prepare the drawing of grillage foundation using Auto cad.
5. To study the commands used in modelling and design of structure using STAAD-Pro.
6. Analysis of 2D Frames.
7. Analysis of 3D Frames.
8. Design of 2D RCC frames and 2D Steel frames according to IS codes.
9. Design of 3D RCC frames and 3D Steel Frames according to IS codes.
10. Design of beams.
11. Analysis of truss frames.

Course Outcomes

At the end of the course, the students will be able to:

- To understand and learn the various codal provisions.
- Ability to prepare 2D and 3 D plans of buildings.
- To efficiently analyze and design of beams, truss frames and staircase.
- To perform various methods of analysis of 2D, 3D frames.

References:

- STAAD Pro Manual.
- IS 456, IS 1893, IS 800, IS 870.

SEMINAR-I						
Category	Seminar					
Course code						
Scheme and Credits	L	T	P	Credits	Semester 6 th	
	0	0	2	1		
External marks:	50 Marks					
Internal marks:	-					
Total	50 Marks					
Duration of Exam	3 Hours					

COURSE CONTENT

During the semester, the students have to:

- Select a topic relevant to analysis, design, implementation, experimental and management of a civil engineering system.
- Undertake a critical review of the literature on the chosen topic.
- Prepare and present a technical report.
- Preparing a review paper for presentation/Publication in national conference/ International conference/ Journals, if possible.

Gurugram University

Scheme of Studies and Examination

B.Tech. (Civil Engineering) – 7th Semester

Sr. No.	Category	Course Code	Course Title	Hours per week			Total Contact Hrs. per week	Credit	Examination Schedule (Marks)				Duration of Exam
				L	T	P			Internal Assessment	Theory	Practical	Total	
1	Professional Core course		Construction planning and management	3	0	0	3	3	30	70	-	100	3
2	Professional Core course		Advanced steel structure	3	0	0	3	3	30	70	-	100	3
3	Professional Elective course		Professional Elective III (Refer List-I)	3	0	0	3	3	30	70	-	100	3
4	Professional Elective course		Professional Elective IV (Refer List-II)	3	0	0	3	3	30	70	-	100	3
5	Open Elective course		Open Elective II	3	0	0	3	3	30	70	-	100	3

6	Open Elective course		Open Elective III	3	0	0	3	3	30	70	-	100	3
7	Professional Core course		Irrigation Drawing LAB	0	0	2	2	1	50	-	50	100	3
8	Mandatory courses (non-credit)		Constitution of India	3	0	0	3	0			GRADE		Refer Note 1*
9	Practical Training		Practical training –II Viva					1	50		50	100	3
10	Project		Industrial Training Synopsis/ Project work-I	0	0	6	4	4	50		50	100	3
		TOTAL						24				900	

***Note: 1.** Constitution of India is a mandatory non-credit course in which the students will be awarded grades A, B, C, F as per their performance. A: Excellent, B: Good, C: Satisfactory, F: Not Satisfactory. A student who is awarded „F“ grade is required to repeat course.

ELECTIVE- II (PEC) –LIST I

S.no	Name of course	Code	L-T-P	Credits
1.	Disaster Management and Mitigation		3-0-0	3
2.	Environmental Management		3-0-0	3
3.	Hydro Power Engineering		3-0-0	3

ELECTIVE- III(PEC) –LIST II

S.no	Name of course	Code	L-T-P	Credits
1.	Ground water engineering		3-0-0	3
2.	Watershed Management		3-0-0	3
3.	River Engineering		3-0-0	3

OPEN ELECTIVE -II (OEC) – LIST III

S.no	Code	Name of course	L-T-P	Credits
1.		Electrical Power Generation	3-0-0	3
2.		Solid and Hazardous waste management	3-0-0	3
3.		Electronic Principles	3-0-0	3

OPEN ELECTIVE- III (OEC) –LIST IV

S.no	Name of course	Code	L-T-P	Credits
1.	Railway and Airport Engineering		3-0-0	3
2.	Traffic Engineering		3-0-0	3
3.	Bridge Engineering		3-0-0	3